

Adaptive TX FFE with Sign-Sign LMS: a Symbol-Rate Simulation on a Backplane Channel

Goal

Reproduce the dual-loop TX FFE adaptation of Stojanović et al. (JSSC 2005) in simulation: a 4-tap transmit FFE adapted by sign-sign LMS from a single receiver-side error sampler, over a real backplane channel (ECEN720 T20 server thru, 10 Gb/s NRZ). The unequalized channel loses 27.6 dB at Nyquist and its worst-case (peak-distortion) eye is closed.

System

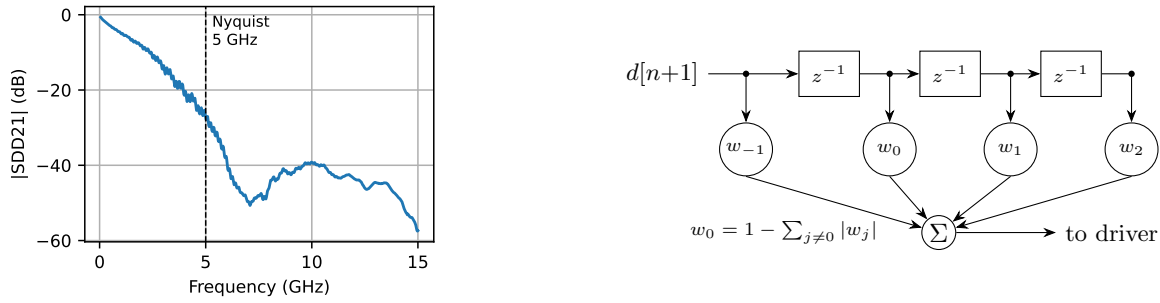


Figure 1: Left: channel loss. Right: symbol-spaced TX FFE (one pre-tap, two post-taps). The TX peak swing is fixed, so the main tap takes whatever the other taps leave: $\sum_j |w_j| = 1$.

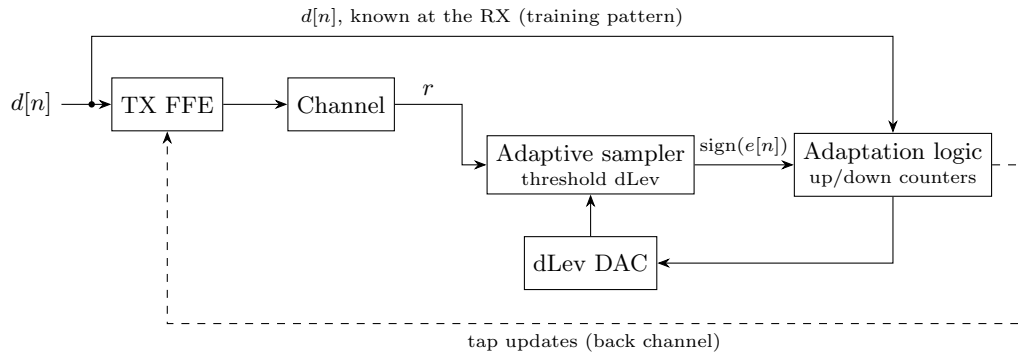


Figure 2: Adaptation loop as simulated. One extra comparator, whose threshold is the dLev DAC, gives $\text{sign}(e)$ directly; no ADC is needed. The data correlated against $\text{sign}(e)$ is the transmitted $d[n]$, i.e. a known training pattern (or error-free RX decisions).

Algorithm

With $e[n] = \text{dLev} - r[n]$, the two loops of the paper are

$$w_j \leftarrow w_j + \Delta_w \text{sign}\left(\sum_n \text{sign}(e[n]) d[n - j]\right), \quad j \neq 0, \quad \text{dLev} \leftarrow \text{dLev} - \Delta_{\text{dLev}} \text{sign}\left(\sum_n \text{sign}(e[n])\right),$$

summed over a block of symbols (one step per block, as an up/down counter would).

- **One sampler.** It sits at +dLev, so only symbols with $d[n] = +1$ contribute.
- **Main tap from the constraint.** Scaling all taps and dLev together leaves every $\text{sign}(e)$ unchanged, so the two loops alone cannot fix the overall scale (and can drift to all-zero). Setting $w_0 = 1 - \sum_{j \neq 0} |w_j|$ removes that freedom. dLev cannot be found first: it is the equalized main cursor, which depends on the final taps.
- **Averaged, it is zero-forcing.** $E[e[n]d[n - j]]$ is the residual ISI at cursor j , so each tap drives its own cursor to zero, even when the unequalized eye is closed.
- **Noise helps.** Without sampler noise the residual ISI takes few discrete values and $\text{sign}(e)$ has a dead zone around the solution (tap error $\sim 5\times$ larger).

Simulation

Symbol-rate model in Python: the channel is its pulse response sampled once per UI at the peak (3 pre-, 40 post-cursors); TX FFE and channel run as a continuous stream across blocks. 3000 blocks $\times$ 512 symbols, $\Delta_w = \Delta_{\text{dLev}} = 10^{-3}$, 5 mV rms sampler noise. The loop uses the transmitted symbols, aligned to each received sample, in place of RX decisions. Eyes are then checked with a waveform-level link (32 samples/UI, full S-parameter channel).

Results

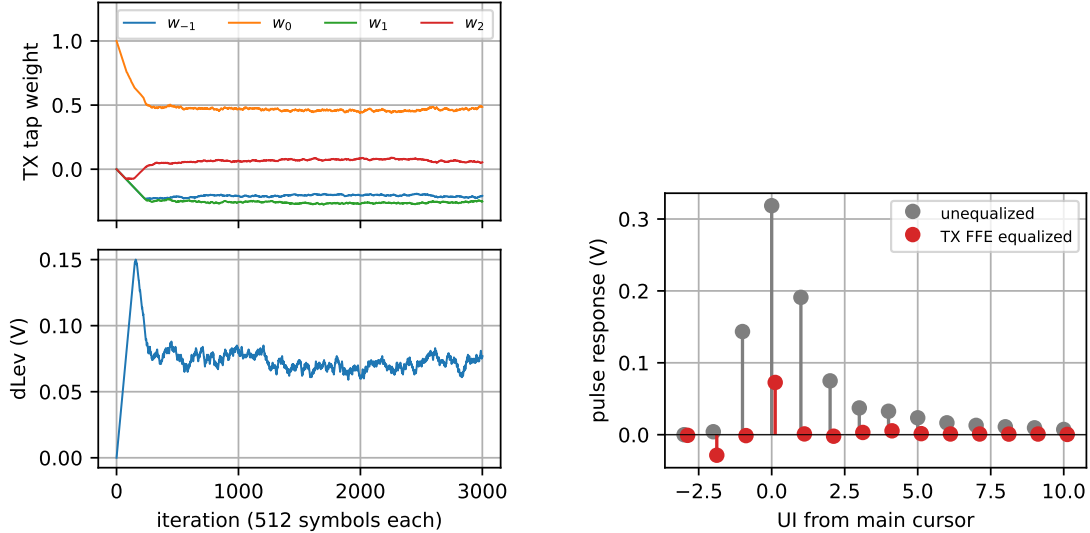


Figure 3: Left: taps and dLev converge together in ~ 300 blocks; dLev overshoots, then follows the main cursor down as pre-emphasis takes swing. Right: UI-spaced pulse response; the cursors within the FFE span (-1 , $+1$, $+2$) are driven to zero.

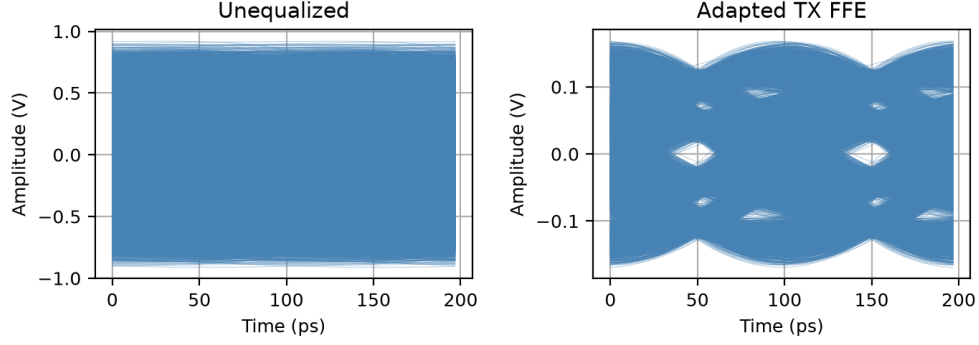


Figure 4: 10 000 random bits through the waveform-level link, before and after adaptation.

	main cursor	$\sum ISI $	PDA eye height
Unequalized	0.319 V	0.630 V	-0.622 V (closed)
Adapted TX FFE	0.073 V	0.061 V	+0.024 V (open)

Table 1: Adapted taps $[w_{-1}, w_0, w_1, w_2] = [-0.212, 0.470, -0.256, 0.061]$ (mean of the last 500 blocks); the exact zero-forcing solution is $[-0.205, 0.462, -0.261, 0.072]$.

Takeaways

- The loop finds the zero-forcing taps blind, from one comparator, and opens a closed eye.
- The cost is swing: the received main cursor drops $4.4\times$, so the eye barely opens. Residual ISI is now outside the FFE span (a new pre-2 cursor and the long post-cursor tail), which points to adding a DFE next (the paper’s loop-unrolled 1-tap DFE).
- At this low SNR, sign-sign keeps wandering slowly around the solution; results are averaged.

Adding a 1-tap DFE

The TX FFE pays for every cursor it cancels with swing: its main tap fell to 0.47, so the received main cursor dropped $4.4\times$. A receiver DFE subtracts the first post-cursor using past decisions instead, at no TX swing cost. Configuration as in the paper: the same 4-tap TX FIR plus a 1-tap DFE, with all loops adapted by sign-sign LMS. (The paper finds the DFE tap by locking dLev to two signal levels; here it is adapted directly.)

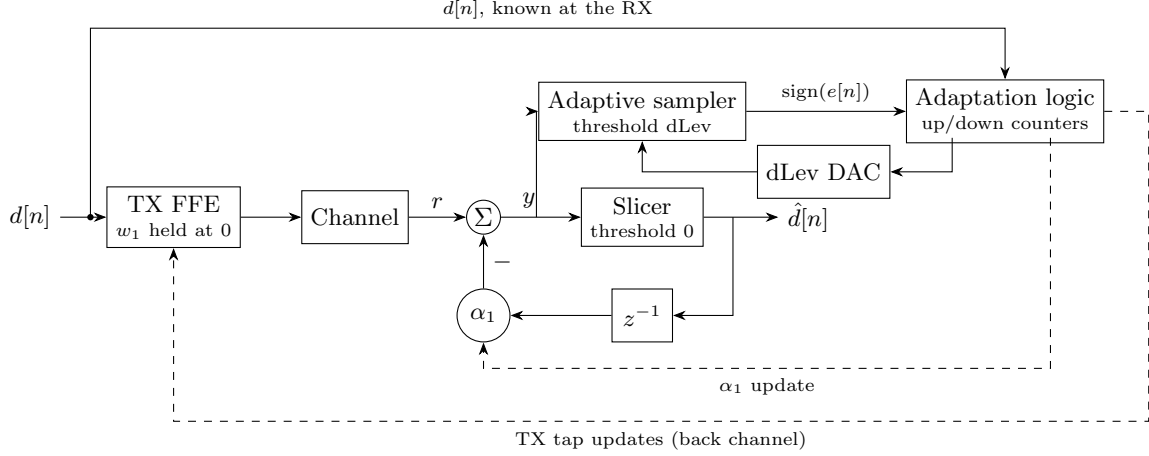


Figure 5: TX FFE + 1-tap DFE. The adaptive sampler sees the DFE summer output y ; the same $\text{sign}(e)$ updates the TX taps, α_1 and dLev (through the dLev DAC).

With $e[n] = \text{dLev} - y[n]$, the DFE tap adds one more counter:

$$\alpha_k \leftarrow \alpha_k - \Delta_\alpha \text{sign}\left(\sum_n \text{sign}(e[n]) d[n - k]\right) \quad (\text{minus: the DFE subtracts}).$$

- **One knob per cursor.** Averaged, w_j drives p_j to 0, α_k tracks p_k , and dLev tracks p_0 . A TX post-tap at a cursor the DFE covers is driven by the same $p_k - \alpha_k$ as α_k , so the split between them would be undetermined: TX post1 is held at 0.
- **All loops together.** The TX equations don't depend on α or dLev; those only track cursors the TX sets, so they follow it as it converges.

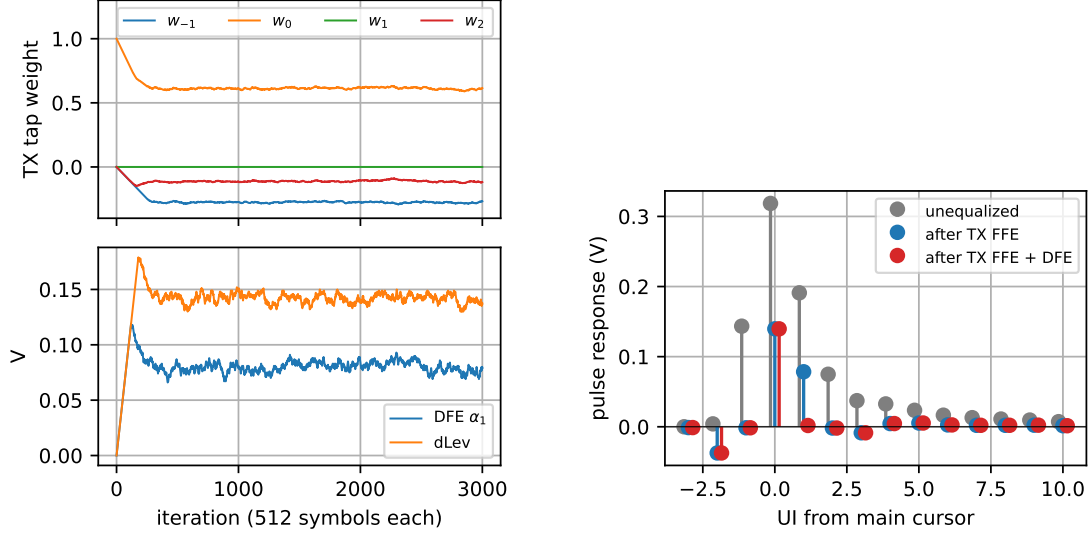


Figure 6: Left: TX post1 (w_1) stays at 0 and the main tap settles near 0.6; α_1 and dLev converge with the TX taps. Right: after the TX FFE the first post-cursor remains (blue); the DFE removes it (red), and the main cursor is about twice the FFE-only one.

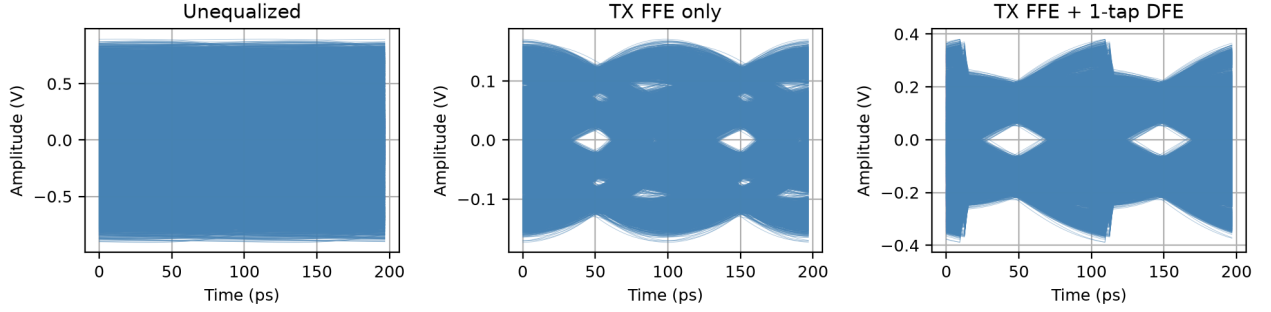


Figure 7: Waveform-level eyes, 10000 random bits. The DFE decides at the pulse-peak instant from its own past decisions; its feedback is held over the UI centered on that instant (hence the steps away from the eye center).

	TX taps $[w_{-1}, w_0, w_1, w_2]$	α_1	main cursor	PDA eye height
Unequalized			0.319 V	-0.622 V
TX FFE only	$[-0.212, 0.470, -0.256, 0.061]$		0.073 V	+0.024 V
TX FFE + 1-tap DFE	$[-0.278, 0.606, 0, -0.116]$	0.077 V	0.140 V	+0.091 V

Table 2: Adapted settings (mean of the last 500 blocks) and the resulting worst-case eye.

Takeaways. The first DFE tap gives most of the gain ($3.8\times$ the PDA eye): it takes the largest post-cursor without spending swing, so the main tap returns to 0.61. The TX still has to cancel the pre-cursor (its pre-tap grows to -0.28) and now also post2. In a sweep, more DFE taps saturate near $+0.12$ V.

Baud-Rate Timing Recovery (Mueller-Müller)

So far the RX sampled exactly at the pulse peak, i.e. an ideal CDR. Mueller and Müller (IEEE Trans. Commun. 1976) recover the sampling phase from one sample per UI, the same samples the adaptation already uses, instead of extra edge samplers at the zero crossings. Their type-A loop locks where the pulse's first pre- and post-cursors are equal, $h_{-1} = h_1$.

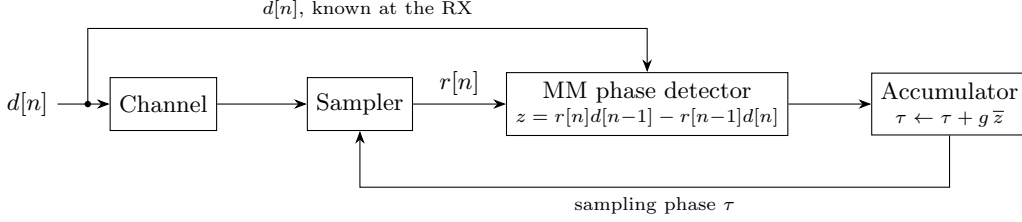


Figure 8: Timing loop as simulated: no equalization, known data, linear (not sign) detector.

Estimating $h_{\pm 1}$ from data. With random data the cursors are never seen alone: at phase τ , $r[n] = \sum_k h_k d[n-k]$ with $h_k = p(\tau + kT)$. For independent ± 1 symbols $E\{d[i]d[j]\} = \delta_{ij}$, so correlating with a neighbouring symbol picks out one cursor:

$$E\{r[n]d[n-1]\} = h_1, \quad E\{r[n-1]d[n]\} = h_{-1}, \quad E\{z[n]\} = h_1 - h_{-1}.$$

$h_1 > h_{-1}$ means sampling early, so the phase moves later by g times z averaged over a block. h_0 cancels in z , so at lock the estimate carries only residual ISI and noise, and it is the same correlation as the sign-sign LMS votes for TX taps ± 1 .

Simulation. The channel keeps its whole pulse response (64 samples/UI) and the cursors are interpolated at the current τ ; a phase change only affects samples taken afterwards. Unequalized T20 channel, transmitted symbols as d , blocks of 1000 symbols, $g = 0.5$ UI/V.

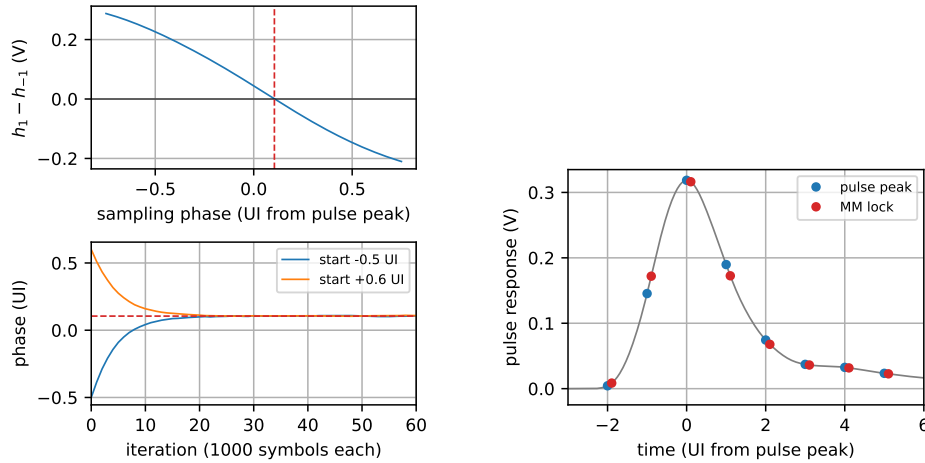


Figure 9: Left: detector characteristic $h_1 - h_{-1}$ (single zero at +0.105 UI) and the loop locking there from either side within ~ 20 blocks. Right: UI-spaced samples at the pulse peak and at the MM lock point.

Takeaways.

- The loop locks at +0.106 UI (rms 0.002, last 100 blocks), where $h_{-1} = h_1 = 0.172$ V: after the peak, because the pulse decays more slowly than it rises. That costs only 1% of h_0 (0.316 V vs. 0.319 V at the peak).
- It converges with the eye fully closed: the estimate needs only independent data, not correct decisions. With its own decisions (14% wrong here) the correlation would be biased.

Sign-Sign Mueller-Müller with a dLev Loop

Without an ADC the RX has only comparators. Slicing r at 0 loses everything, $d[n]d[n-1] - d[n-1]d[n] = 0$. Subtracting the main cursor first keeps it: with $e[n] = r[n] - h_0 d[n]$ the h_0 terms cancel exactly, $r[n]d[n-1] - r[n-1]d[n] = e[n]d[n-1] - e[n-1]d[n]$. Two error samplers at $\pm dLev$ and an XOR (Spagna, ISSCC 2010) give, for correct decisions and dLev in place of h_0 ,

$$\begin{aligned} \text{ERR}[n] &= \text{sign}(|r[n]| - dLev) = d[n] \text{sign}(e[n]), \\ \text{sign}(e[n]) d[n-1] - \text{sign}(e[n-1]) d[n] &= d[n]d[n-1](\text{ERR}[n] - \text{ERR}[n-1]), \end{aligned}$$

so only consecutive ERRs that differ vote. The same ERRs drive dLev, one step per block:

$$\tau \leftarrow \tau + \Delta_\tau \text{sign}\left(\sum_n d[n]d[n-1](\text{ERR}[n] - \text{ERR}[n-1])\right), \quad dLev \leftarrow dLev + \Delta_{dLev} \text{sign}\left(\sum_n \text{ERR}[n]\right).$$

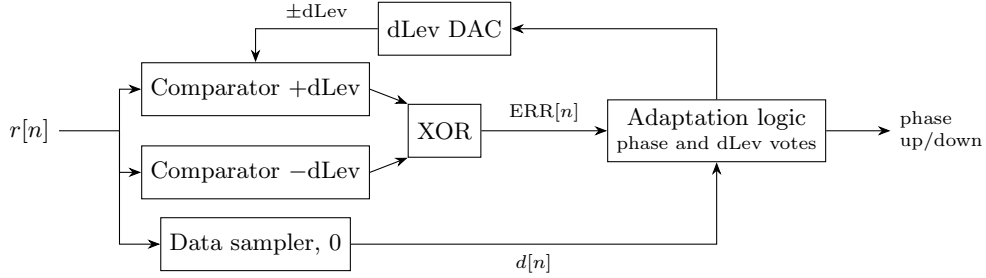


Figure 10: Comparator-only receiver as in Spagna (ISSCC 2010); the DAC sets both error thresholds. In the simulation the transmitted $d[n]$ selects the error sampler at $d[n] \cdot dLev$, which equals the XOR form for correct decisions.

The lock point does not depend on dLev. With $\delta = h_0 - dLev$ and R the ISI from $|k| \geq 2$, $E\{\text{sign}(e[n]) d[n-1]\} = G(h_1, h_{-1})$ and $E\{\text{sign}(e[n-1]) d[n]\} = G(h_{-1}, h_1)$, where $G(a, b) = E\{\text{sign}(\delta x + ay + bw + R)y\}$ for independent ± 1 symbols x, y, w (R has the same distribution in both). At $h_1 = h_{-1}$ the two are equal for any δ , so dLev sets only the detector's gain. dLev itself settles at the median of $r[n]d[n]$, which is h_0 because the ISI is symmetric about 0.

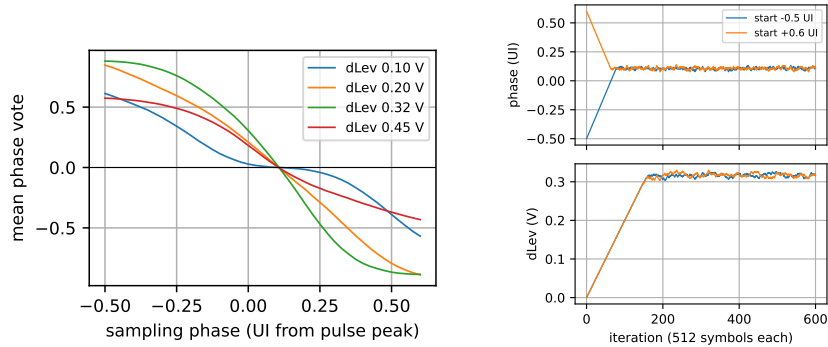


Figure 11: T20, unequalized, known data, 5 mV rms comparator noise. Left: mean phase vote at fixed dLev; every dLev crosses zero at the same phase, only the slope changes. Right: the loops from $dLev = 0$, blocks of 512 symbols, $\Delta_\tau = 1/128$ UI, $\Delta_{dLev} = 2$ mV; the phase locks before dLev arrives.

Takeaways.

- The phase locks at $+0.107$ UI, as the linear loop does ($+0.105$ UI), in ~ 80 blocks while dLev is still below half of h_0 ; dLev then settles at 0.316 V, the h_0 there. The loops do not fight: dLev only scales the detector gain.
- Sign-sign dithers around the lock (rms 0.01 UI vs. 0.002 linear). With few discrete ISI values and dLev near h_0 the gain can vanish; T20's long tail smooths the ISI.

Recovering the Training Data: PRBS Sync

Every loop so far was given the transmitted $d[n]$. That is how link training works (the TX sends a known PRBS), but the RX still has to find where in the PRBS each sample is. A PRBS comes from a two-tap LFSR, $b[n] = b[n - N] \oplus b[n - t]$ for PRBSN (PRBS15: $t = 14$), so any N consecutive bits determine everything after them. The RX locks onto it from its own decisions:

- **Seed and verify.** N decisions become the LFSR state; the next 256 decisions are predicted from it. A right seed disagrees only where the decisions are wrong, a wrong one $\sim 50\%$ of the time: lock if at most 30% disagree.
- **Retry one bit later.** On failure the seed slides by one bit, re-checked against buffered decisions. Seeding from the latest decisions instead cycled through 6 fixed positions of PRBS7 on T20 and never locked: without noise the decision errors are set by ISI and repeat with the PRBS period, and none of the 33 error-free windows was ever tried.
- **Free-run.** Once locked, the LFSR no longer reads the decisions: the reference is error-free however closed the eye is.

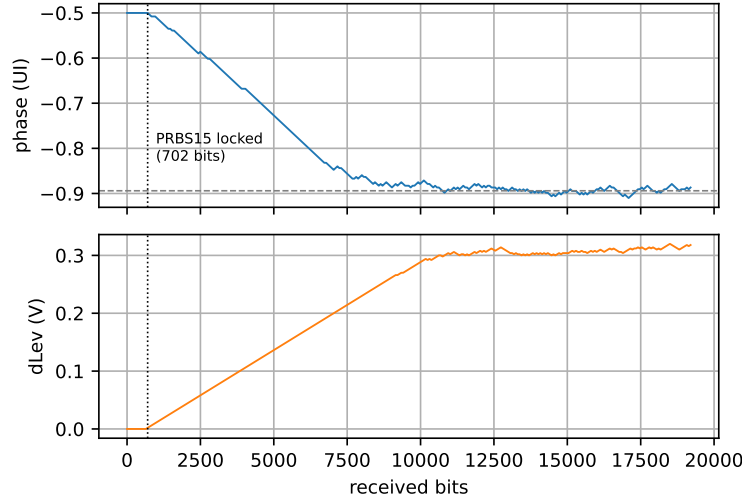


Figure 12: T20, unequalized, starting at -0.5 UI (the worst phase, 25% decision errors). The SS-MM CDR and dLev hold until the PRBS15 sync locks after 702 bits, then converge on the recovered d : -0.894 UI and 0.308 V. Dashed: the known-data lock point $+0.106$ UI, one UI over.

Takeaways.

- With 14–26% decision errors, the sync locks within 270–480 bits for PRBS7 and PRBS15 at any sampling phase, and the recovered d then has no errors.
- The sync may label the data one bit off; the RX cannot tell and need not. Here it locks to $d[n - 1]$, so the CDR settles one UI earlier on the same point of the pulse: $-0.894 = 0.106 - 1$, with the same dLev, and the recovered d equals the transmitted $d[n - 1]$ exactly.
- Long PRBSs can false-lock on a closed eye. A seed with one wrong bit diverges from the truth only as the error propagates through the taps, N bits per step:

disagreement of a one-bit-off seed	PRBS7	PRBS15	PRBS31
over 256 bits	50%	32%	12%
over 4096 bits	50%	48%	34%

A one-bit-off PRBS31 seed passes the 30% test, and the CDR then trains on wrong data (dLev stayed near 0). Use a short PRBS, or a verification window many times N .